SELECTED ENGINEERING PROJECTS

AI Automation & Intelligent Workflow Engineering Portfolio

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ENGINEERING PHILOSOPHY

My approach to automation begins with understanding the operational system rather than selecting automation tools. Every solution is designed by analysing how information moves through an organisation, identifying structural bottlenecks, and engineering reliable workflows that improve operational performance while remaining maintainable over time.

Instead of automating isolated tasks, I architect complete workflow systems that integrate business rules, APIs, AI-assisted decision support, monitoring, reporting, and governance into a cohesive operational platform.

- **Reliability before complexity**
- **Data integrity before automation speed**
- **Business objectives before technology selection**
- **Modularity before customisation**
- **Operational visibility before optimisation**

CORE TECHNICAL CAPABILITIES

WORKFLOW ENGINEERING

n8n · Workflow Orchestration · Business Process Automation

AI

OpenAI · Claude · Gemini · Prompt Engineering

PROGRAMMING

JavaScript · Python

INTEGRATION

REST APIs · Webhooks · Google Workspace

ENGINEERING

Solution Architecture · Modular Design · Error Handling · Monitoring · Audit Logging · Workflow Governance

WHAT'S INSIDE

PROJECT ONE

AI-Powered Patient Workflow Automation Platform (Meridux PatientCore)

Business Challenge

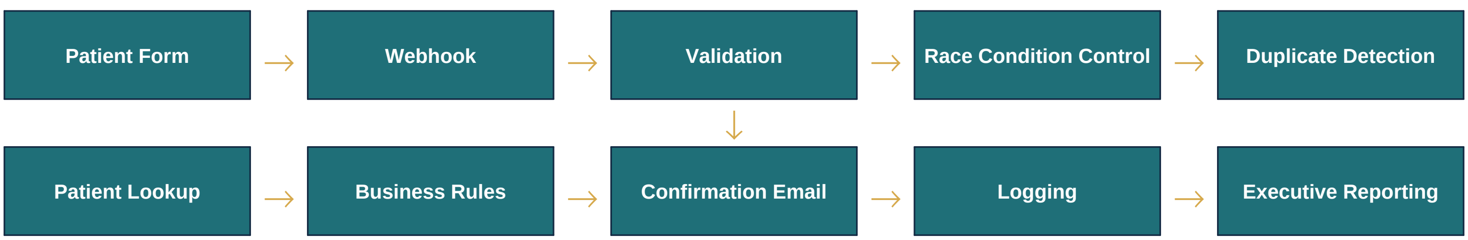
Healthcare providers often depend on manual administrative processes for handling patient enquiries, appointment requests, confirmations, duplicate checking, reporting, and follow-up activities. These fragmented workflows increase staff workload, introduce avoidable errors, delay response times, and reduce operational visibility.

The objective was to engineer a workflow architecture capable of automating the complete patient intake process while maintaining data accuracy, reliability, and operational transparency.

Design Objectives

- Eliminate repetitive administrative work
- Improve data consistency
- Prevent duplicate submissions
- Automate patient communication
- Generate operational reporting
- Build reusable workflow components
- Create an architecture suitable for future expansion

System Architecture



Engineering Decisions

Rather than storing workflow state within multiple disconnected applications, the architecture centralises operational data while separating validation, routing, reporting, and communication into independent workflow components.

This modular structure improves maintainability and enables future enhancements without redesigning the complete system.

PROJECT ONE (CONTINUED)

Engineering Highlights

Workflow Orchestration

Designed a ten-stage workflow that automates patient processing from initial submission through confirmation and reporting.

Race-Condition Prevention

Engineered concurrency controls that prevent simultaneous submissions from generating duplicate workflow execution.

Intelligent Routing

Business rules automatically distinguish between new and returning patients, enabling conditional workflow execution.

Error Handling

Structured validation and monitoring ensure incomplete or inconsistent submissions are safely isolated without interrupting the primary workflow.

Operational Reporting

Daily executive summaries provide management with accurate operational insight without requiring manual reporting activities.

Technology Stack

n8n	JavaScript	REST APIs	Google Workspace
Google Sheets	Gmail API	OpenAI	Webhooks

Business Impact

The completed platform demonstrates how engineering principles can replace fragmented manual administration with an intelligent operational system that improves efficiency, reporting, and long-term maintainability.

PROJECT TWO

Prospect Intelligence Engine (PIE)

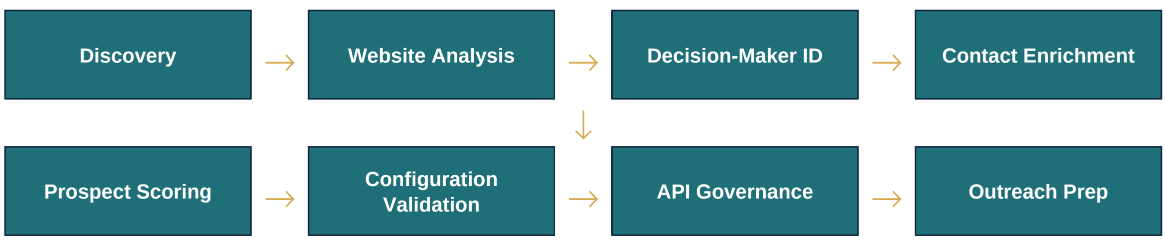
Business Challenge

Sales teams frequently rely on fragmented manual research when identifying prospects, collecting contact information, evaluating opportunity quality, and preparing outreach activities. This manual approach limits scalability and introduces inconsistencies throughout the prospecting process.

Design Objectives

- Automate business discovery
- Standardise prospect evaluation
- Improve data quality
- Reduce repetitive research
- Build reusable automation modules
- Support future expansion

Architectural Overview



Engineering Highlights

Modular Architecture
Each functional capability operates independently while contributing to a unified automation platform.

Configuration-Driven Design
Workflow behaviour is controlled through structured configuration rather than embedded logic, simplifying maintenance.

API Governance
Quota monitoring prevents service interruption while improving workflow reliability.

Data Integrity
Duplicate prevention and validation improve information quality before downstream processing.

Scalability
The architecture supports the addition of future workflow modules without affecting existing functionality.

Technology Stack

- n8n
- Google Places API
- REST APIs
- JavaScript
- Google Sheets

ENGINEERING PRINCIPLES

Designing Reliable AI Automation Systems

Every workflow I design follows a consistent engineering methodology.

1 **Understand the System Before the Technology**
Automation should be based on operational requirements rather than available software tools.

2 **Design for Reliability**
Every workflow should continue operating predictably under normal and exceptional conditions.

3 **Preserve Data Integrity**
Validation, duplicate prevention, concurrency control, and audit logging protect the quality of operational data.

4 **Build Modular Architectures**
Reusable workflow components simplify maintenance and accelerate future development.

5 **Prioritise Operational Visibility**
Automated reporting, monitoring, and structured logging enable organisations to understand system performance without manual investigation.

6 **Engineer for Business Outcomes**
Technology is valuable only when it improves efficiency, reliability, decision-making, or customer experience.

> *My work combines systems engineering, software integration, and AI-powered workflow automation to transform complex operational processes into reliable, scalable business systems. I approach every solution as an architectural challenge, ensuring that technology serves clearly defined business objectives rather than becoming an objective in itself.*